THE MATHEMATICS IN GROTHENDIECK'S LAST PAPERS: A FIRST DESCRIPTION OF THE BNF FONDS NAF 29093, 1991–2000

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Abstract. The papers Grothendieck wrote at Lasserre after 1987 entered the Bibliothèque nationale de France in 2023 as NAF 29093: forty-three boxes and about 22 500 leaves, almost all of them filed under *Réflexions sur la Vie et le Cosmos*, with a single volume presented as mathematical. They can be read only in the reading room. This note describes what three sessions there, in October and November 2026, found of mathematics in seven boxes. The 1992 volume, *Géométrie élémentaire schématique*, is a continuous text of 549 leaves in the style of EGA on invertible modules, quadratic forms, discriminants and quadratic groups over a base; its subjects are those of two paragraphs announced in the 1965 plan of EGA IV and never published, and the author calls it a chapter of the EGA. It is followed by a course on triality (1995), notes on a combinatorial geography (1996) and dated mathematics on paths on a chessboard and a cube (1997–1998). The physics of *Esprit et Matière* and of the tenth part of *Le Problème du Mal* uses classical mathematics, not the notions of his last mathematical texts. None of the boxes read returns to motives, to Teichmüller theory, to the geometry of forms of 1986 or to the pseudolines of the Montpellier notes.

1. The fonds

Grothendieck's mathematical working notes up to August 1991 are at the University of Montpellier, which publishes their facsimiles. The papers of the following years, written at Lasserre, entered the Bibliothèque nationale de France in 2023, bequeathed by their author and completed by a gift of his children, and are described by the Département des Manuscrits as NAF 29093, *Fonds Alexandre Grothendieck. uvres.*, 1987–1999, forty-three boxes [3, 2]. The notices of the boxes count 22 495 leaves, of which 15 274 belong to *Le Problème du Mal*. The boxes consulted are digitised but are served only in the reading room, after a request to the department; photographs were not allowed.

The inventory files almost everything under *Réflexions sur la Vie et le Cosmos*: *La Clef des Songes* (boxes 1–2, 1987–1988), *Structures de la Psyché* and *Psyché et Structure* (4–7), *Le Problème du Mal* (8–13, 16–33, 35–39, 1993–1998), *Esprit et Matière* (14–15, 34, 1992–1996), *Réflexions sur la Création* (41), the two « Gribouillis » (40, 42) and a box of notes and scribbles from August 1991 to February 1997 (43). One volume only is presented as mathematical, the *Géométrie élémentaire schématique* (box 3).

The library's rule is that no transcription of these papers may be put online without the authorisation of the heirs, and that very short quotations are allowed. What follows describes; it quotes only titles and a few words.

2. How the boxes were read

The boxes were requested in an order fixed by questions prepared from the Montpellier notes: did he return to motives after 1991, do the threads of the Montpellier folders go on, and what is the mathematics of the volume presented as such. Three sessions, on 23 and 29 October and 6 November 2026, read seven boxes, pochette by pochette, with a sheet of questions box by box [6]. For each pochette the reader noted dates in his hand, the share

Date: 7 November 2026.

2020 *Mathematics Subject Classification.* 01A60; 14-03, 11E81.

Key words and phrases. Grothendieck, Lasserre, Bibliothèque nationale de France, manuscripts, quadratic forms, triality, EGA.

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of mathematics (an estimate by eye), the subjects, the names cited and the answers to the questions. Table 1 summarises them.

Box	Title (inventory)	Dates	Math.	What the pochettes read show
3	Géométrie élémentaire schématique	19.8–1.10.1992	100%	continuous EGA-style text, 18 sections; stops in § 18.8
43	notes and scribbles, 1991–1997	1995–1996	mixed	*Trialité* (Nov. 1995); combinatorial geography (June 1996); physics notes
40	Gribouillis I	24.10.1997–3.2.1998	95%	paths on a chessboard and a cube; symmetric groups; « bitrialité projective »
14	Esprit et Matière, A–C	1993–1996	5–80%	Maxwell and Liénard–Wiechert; an axiomatics of cosmic space; Ricci curvature
34	Esprit et Matière, E	1992–1996*	10%	physics calculations; powers of 3
35	Le Problème du Mal X, IX–XIII	1996–1997*	20%	Planck's law marked in blue (f. 887)
39	Le Problème du Mal, tables	14.6.1998–4.1.2000	30%	a physics text under the name of a table

TABLE 1. The boxes read in October and November 2026: the dates read in his hand in the pochettes consulted (with *, the notice's dates), and an estimate by eye of the share of mathematics. Boxes 15, 41 and 42 are for the next session.

3. The 1992 volume: a chapter of EGA

The *Géométrie élémentaire schématique* is entirely mathematical and does not mention physics; the general tables of the *Réflexions* drawn up by Grothendieck (box 39) do not mention it, although the inventory describes it as auxiliary results for the Riemann–Lorentz structure. Its 549 leaves, written in ink on both sides of ordinary white paper, form eighteen sections dated session by session from 19 August to 1 October 1992, about a dozen leaves a day. They treat invertible modules, bilinear and quadratic forms, discriminants, smooth relative quadrics, projective involutions and quadratic groups over a base S. The text runs on with almost no crossings-out, reread with a felt pen, and its statements are numbered on three levels as in EGA (Proposition 15.1.17, f. 168v). The table announces a last section on quadratic groups and cross-ratios, but the text stops in § 18.8, « Dualités » (f. 517v); of the cross-ratios only computations among the scribbles remain. The other thirty-two leaves of the box are such scribbles.

These subjects are those of two paragraphs that the 1965 plan of EGA IV announced and that never appeared, on smooth forms and ordinary quadratic singularities, and on resultant and discriminant; the plan is in the Montpellier notes (folder 25, p. 134). Grothendieck had drafted part of them for EGA V: typescripts kept at the IHÉS treat smooth quadratic forms over any base, with a corrected discriminant, half the usual one in odd rank, invertible exactly when the form is smooth, and define the resultant and the discriminant of a form of degree n over any base [5]. The subject returns in the Montpellier folders: folder 77 (from 1977) defines the half-discriminant of a quadratic form of odd rank over any base, and folder 84 (around 1982–1986) treats quadratic algebras and quadratic covers over any base without dividing by 2. The half-discriminant statement of folder 77 is among those proved in Lean in the companion paper [4]. The 1992 text takes these subjects up again in the language of EGA, without reference to any volume; on f. 378 he describes what he is writing as a very large chapter of the EGA, and he remarks on a point of terminology he had not thought of at the

time of EGA and SGA (f. 156v). Seven years earlier, in *Récoltes et semailles*, he spoke of a later chapter of the EGA only as one that never saw the light of day [7].

The case recalls that of Ramanujan's « lost » notebook, found by Andrews in 1976 among Watson's papers at Trinity College: leaves of a last period, kept in a fonds and unread for half a century [1].

4. Triality, geography, a chessboard

Box 43 holds *Trialité (vectorielle et projective)*, eighty-nine leaves written as a course from 8 to 20 November 1995, on the same paper and with the same care. Its stated motivation is a canonical abelian decomposition of physical space, identified with the real projective space $\mathbf{P}^3(\mathbf{R})$. Its fifteen sections go from ε-bases of a module to vector and projective trialities, a combinatorial axiomatics of triality, bitrialities and triality over $\mathbf{F}_3$; the statements are numbered and dated, at about 170 words a page, and a submodule with its ε corresponds to a set with a fixed-point-free involution.

The same box holds notes on a « géographie combinatoire » (June 1996): finite « forestiers » sets whose elements play the part of geographical sectors. The sectors nest along a forest whose minimal vertices are localities, and each sector has for inhabitants the disjoint union of those of the localities it contains; a cube crossed by an ordered path appears. The vocabulary is not new, since the object axiomatised in *Vers une géométrie des formes* (1986) is called a « contrée » and the graphs of the Teichmüller notes are geographical representations (folder 146), but the notion is another, and the words of 1986 do not occur in the leaves read.

The first « Gribouillis » (box 40, October 1997 to February 1998) are, despite their title, continued statements and proofs dated to the day (24 October and 15 November 1997, 9, 15 and 27 January, 1 and 3 February 1998), with only a few pages of draft computation, about 95% mathematics. They study moves on a chessboard that do not return to a square, with sub-squares of size 3, and paths on a cube, with the symmetric groups that act on them: $\mathfrak{S}_4$, plausibly the rotation group of the cube acting on its diagonals, $\mathfrak{S}_I$, $\mathfrak{S}_\rho$, and maps $\mathfrak{S} \to \mathfrak{S}$ and $\mathfrak{S} \to \mathrm{Aut}(\mathfrak{S})$. A « bitrialité projective » is noted on 3 February 1998; an algorithm named after Azumaya is mentioned. There is no reference to the 1992 volume or to *Trialité*, and no return of Teichmüller theory, of $M_{g,\nu}$, of dessins or of $\mathrm{Gal}(\bar{\mathbf{Q}}/\mathbf{Q})$.

5. The mathematics of the physics

The physics, which the inventory describes as a radical critique of modern physics, uses classical mathematics. In box 14 (*Esprit et Matière*, sections A–C), the section « Maxwell II » revises his own earlier formulas in the light of those of Liénard and Wiechert, passes to the relativistic form with differential forms, and uses the Lorentz quadratic form, without defining it, to derive the wave equation of the potential; a margin note dated 3 January 1994 corrects a forgotten factor $1/c^2$ (f. 177v). It ends with a property he calls global ML, given by two conservation equations, and the decomposition of an open set into a product of three factors. Section B, dated from 25 October 1993, sets an axiomatics (A1)–(A19) of cosmic space in a continuous Riemannian frame, about 5% mathematics. Section C, preparatory sheets, computes the Ricci curvature of a metric with a radial variable and states the existence and uniqueness of a torsion-free connection for which the metric is parallel, the Levi-Civita connection, in his words. In box 34 (section E), the chapter titled « Le Champ est une trialité » puts physics calculations in fair copy without referring to the triality of 1995: the word there names the triad of gravitation, electromagnetism and radiation. A precise bridge does appear in the chapters on the Moon in the same box, where the field is written $E+iB$ and a quadratic form is defined by the exterior product, with the magnetic field dual to the electric one (f. 402): the algebra of quadratic forms and exterior products of the 1992 volume, applied to Maxwell's equations. The same box shows his arithmetic of powers of 3: $3^6 - 1 = 26 \cdot 28$ at f. 498, with $28 = 3^3 + 1$ joined to the lunar period.

In the tenth part of *Le Problème du Mal* (box 35), the chapter titled after « la formule la plus belle » marks with a blue line in the margin, as he marks an important result, Planck's

law for the spectral radiance (f. 887); the names cited are Feynman, Landau, de Broglie and Kastler. Its « table » in box 39 is in fact a developed physics text of 290 leaves, dated from 14 June 1998 to 4 January 2000, after the end date of the fonds as catalogued. A sheet of box 43 lists what he took for the errors of established physics, on the conservation of energy, the emission of photons and the Poynting vector.

The project is old: in 1986 he wrote to Yamashita that space-time is probably finite and the mathematical continuum only a convenient fiction, as Riemann would have said [8]. In the boxes read, the physics does not use his own notions of 1986–1996: no places or figures without points, no pseudolines, no signs on triples, no finite axiomatics of space.

6. Paper and dates

The paper is reused as in Montpellier: many sheets carry the letterhead of an estate agency; two sheets of fan-fold printer paper, printed with the output of a finite-element program in Fortran, serve as a wrapper for a group of pages, as reused listings do in folder 154. Such paper gives only a lower bound for the date of what is written on it. Part of the third pochette of box 14 is written on the back of a letter later than February 1996, outside the period 1992–1993 the notice gives for the box; box 39 runs to January 2000. The notices' dates are therefore those of the main texts, not of every sheet.

7. What remains

The question that led to the visit, whether he returned to motives after 1991, has no positive answer in the seven boxes read; the boxes of *Le Problème du Mal* not read, and boxes 15, 41 and 42, may still change that. The next session is planned for section D of *Esprit et Matière* (box 15) and a survey of the second « Gribouillis » (box 42), to find the last dated mathematics. A fuller account would need the heirs' authorisation to quote, and a reading of *Trialité* and of the 1992 volume by a specialist of quadratic forms over a base, whose subject is classical but whose text has had no reader.

Use of language models. The questions for the visits were prepared, and the notes taken in the reading room were organised and checked against the inventory, with language models (Claude Opus 5.5, Anthropic); the boxes were read by the author, in person, and the readings reported here are his. A first draft of this text was produced by the same models from those notes, under the author's direction.

Acknowledgements. The Département des Manuscrits of the Bibliothèque nationale de France served the boxes.

References

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